

Calculus I

Homework

Limits

1. Evaluate the difference quotient and simplify your answer.

(a) $\frac{f(2+h) - f(2)}{h}$, where $f(x) = x^2 - x - 1$.

ANSWER: $h + 3$

(b) $\frac{f(a+h) - f(a)}{h}$, where $f(x) = x^2$.

ANSWER: $h + 2a$

(c) $\frac{f(a+h) - f(a)}{h}$, where $f(x) = x^3$.

ANSWER: $h^2 + 3a^2 + 3ah$

(d) $\frac{f(a+h) - f(a)}{h}$, where $f(x) = x^4$.

ANSWER: $6a^3h + 4a^2h^2 + 3ah^3 + h^4$

(e) $\frac{f(x) - f(a)}{x - a}$, where $f(x) = \frac{1}{x}$.

ANSWER: $-\frac{1}{x^2}$

(f) $\frac{f(x) - f(1)}{x - 1}$, where $f(x) = \frac{x-1}{x+1}$.

ANSWER: $\frac{x}{1+x^2}$

2. Write down a formula for a function whose graphs is given below. The graphs are up to scale. Please note that there is more than one way to write down a correct answer.

$$\text{answer: } y = \begin{cases} -\frac{3}{2}x - \frac{1}{2} & \text{if } 2 \leq x < 3 \\ -\frac{3}{2}x - 1 & \text{if } 1 < x < 2 \end{cases}$$

$$\text{answer: } y = \begin{cases} -\frac{3}{2}x + 3 & \text{if } 0 < x \leq 2 \\ -\frac{3}{2}x - \frac{3}{2} & \text{if } 2 < x \leq 3 \end{cases}$$

(b)

(d)

$$\text{answer: } y = \begin{cases} -3x - 4 & \text{if } -2 \leq x < -1 \\ -2x + 1 & \text{if } -1 \leq x < 0 \\ 3x - 4 & \text{if } 0 \leq x < 1 \end{cases}$$

$$\text{answer: } y = \begin{cases} -\frac{3}{2}x + \frac{3}{2} & \text{if } -3 \leq x < -1 \\ -4x + 6 & \text{if } -1 < x \leq 2 \end{cases}$$

(c)

3. Write down formulas for function whose graphs are as follows. The graphs are up to scale. All arcs are parts of circles.

(a)

4. Evaluate the difference quotient and simplify your answer.

(a) $\frac{f(2+h) - f(2)}{h}$, where $f(x) = x^2 - x - 1$.

(d) $\frac{f(a+h) - f(a)}{h}$, where $f(x) = x^4$.

(b) $\frac{f(a+h) - f(a)}{h}$, where $f(x) = x^2$.

(e) $\frac{f(x) - f(a)}{x - a}$, where $f(x) = \frac{1}{x}$.

(c) $\frac{f(a+h) - f(a)}{h}$, where $f(x) = x^3$.

(f) $\frac{f(x) - f(1)}{x - 1}$, where $f(x) = \frac{x-1}{x+1}$.

5. Find the implied domain of the function.

(a) $f(x) = \frac{x+4}{x^2-4}$.

(e) $h(x) = \frac{1}{\sqrt[6]{x^2-7x}}$.

(b) $f(x) = \frac{2x^3-5}{x^2+5x+6}$.

(f) $f(u) = \frac{u+1}{1+\frac{1}{u+1}}$.

(c) $f(t) = \sqrt[3]{3t-1}$.

(g) $F(x) = \sqrt{10-\sqrt{x}}$.

(d) $g(t) = \sqrt{5-t} - \sqrt{1+t}$.

6. Find the implied domain of the function.

(a) $f(x) = \frac{x+4}{x^2-4}$.

(e) $h(x) = \frac{1}{\sqrt[6]{x^2-7x}}$.

(b) $f(x) = \frac{2x^3-5}{x^2+5x+6}$.

(f) $f(u) = \frac{u+1}{1+\frac{1}{u+1}}$.

(c) $f(t) = \sqrt[3]{3t-1}$.

(g) $F(x) = \sqrt{10-\sqrt{x}}$.

(d) $g(t) = \sqrt{5-t} - \sqrt{1+t}$.

7. Compute the composite functions $(f \circ g)(x)$, $(g \circ f)(x)$. Simplify your answer to a single fraction. Find the domain of the composite function.

(a) $f(x) = \frac{x+2}{x-2}$, $g(x) = \frac{x-1}{x+2}$.

(b) $f(x) = \frac{x+1}{3x-2}$, $g(x) = \frac{x-2}{x-1}$.

$$\text{answer: } (f \circ g)(x) = \frac{x-5}{3x-2}, \quad (g \circ f)(x) = \frac{x-5}{3x-2}$$

$$\text{answer: } (f \circ g)(x) = \frac{x-5}{3x-2}, \quad (g \circ f)(x) = \frac{x-5}{3x-2}$$

$$(c) f(x) = \frac{2x+1}{3x-1}, g(x) = \frac{x-2}{2x-1}.$$

$$(d) f(x) = \frac{x+1}{x-2}, g(x) = \frac{x+2}{2x-1}.$$

$$(e) f(x) = \frac{5x+1}{4x-1}, g(x) = \frac{4x-1}{3x+1}.$$

$$(f) f(x) = \frac{3x-5}{x-2}, g(x) = \frac{x-2}{x-4}.$$

$$(g) f(x) = \frac{x-3}{x+2}, g(y) = \frac{y+3}{y-4}.$$

$$\text{ANSWER: } \begin{aligned} (f \circ g)(x) &= \frac{-5+4x}{-5+4x} = 1 \\ (f \circ f)(x) &= \frac{x+6}{x-3} \end{aligned}$$

$$\text{ANSWER: } \begin{aligned} (f \circ g)(x) &= \frac{1+3x}{1+3x} = 1 \\ (g \circ f)(x) &= \frac{x+4}{x-4} \end{aligned}$$

$$\text{ANSWER: } \begin{aligned} (f \circ g)(x) &= \frac{-4+23x}{-4+23x} = 1 \\ (f \circ f)(x) &= \frac{1}{x} \end{aligned}$$

$$\text{ANSWER: } \begin{aligned} (f \circ g)(x) &= \frac{-2x+14}{-2x+14} = 1 \\ (g \circ f)(x) &= \frac{x+3}{x-4} \end{aligned}$$

$$\text{ANSWER: } \begin{aligned} (f \circ g)(x) &= \frac{-2x+15}{-2x+15} = 1 \\ (f \circ f)(x) &= \frac{1}{x} \end{aligned}$$

8. Find the functions $f \circ g$, $g \circ f$, $f \circ f$ and $g \circ g$ and their implied domains. The answer key has not been proofread, use with caution.

$$(a) f(x) = x^2 + 1, g(x) = x + 1.$$

$$(b) f(x) = \sqrt{x+1}, g(x) = x + 1.$$

$$(c) f(x) = 2x, g(x) = \tan x.$$

In this subproblem, you are not required to find the domain.

$$(d) f(x) = \frac{x+1}{x-1}, g(x) = \frac{x-1}{x+1}.$$

$$\text{ANSWER: } \begin{aligned} \text{Domain of } f \circ g \text{ is } x > -2. \text{ Domain of } g \circ f \text{ is } x \geq -1. \text{ Domain of } f \circ f \text{ is all reals } (x \in \mathbb{R}). \\ \text{Domain of } g \circ g \text{ is } x \neq \frac{\pi}{2} \text{ for all } k \in \mathbb{Z}. \end{aligned}$$

9. Convert from degrees to radians.

$$(a) 15^\circ.$$

$$(h) 120^\circ.$$

$$(n) 305^\circ.$$

$$(b) 30^\circ.$$

$$(i) 135^\circ.$$

$$(o) 360^\circ.$$

$$(c) 36^\circ.$$

$$(j) 150^\circ.$$

$$(p) 405^\circ.$$

$$(d) 45^\circ.$$

$$(k) 180^\circ.$$

$$(q) 1200^\circ.$$

$$(e) 60^\circ.$$

$$(l) 225^\circ.$$

$$(r) -900^\circ.$$

$$(f) 75^\circ.$$

$$(m) 270^\circ.$$

$$(s) -2014^\circ.$$

$$(g) 90^\circ.$$

10. Convert from radians to degrees. The answer key has not been proofread, use with caution.

$$(a) 4\pi.$$

$$(c) \frac{7}{12}\pi.$$

$$(e) -\frac{3}{8}\pi.$$

$$(b) -\frac{7}{6}\pi.$$

$$(d) \frac{4}{3}\pi.$$

$$(f) 2014\pi.$$

(g) 5.

(h) -2014.

$$\frac{1}{1000} \approx \frac{1}{1000}$$

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11. Prove the trigonometry identities.

(a) $\sin \theta \cot \theta = \cos \theta$.

(b) $(\sin \theta + \cos \theta)^2 = 1 + \sin(2\theta)$.

(c) $\sec \theta - \cos \theta = \tan \theta \sin \theta$.

(d) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$.

(e) $\cot^2 \theta + \sec^2 \theta = \tan^2 \theta + \csc^2 \theta$.

(f) $2 \csc(2\theta) = \sec \theta \csc \theta$.

(g) $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$.

(h) $\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 2 \sec^2 \theta$.

(i) $\tan \alpha + \tan \beta = \frac{\sin(\alpha + \beta)}{\cos \alpha \cos \beta}$.

(j) $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$.

(k) $\sin(3\theta) + \sin \theta = 2 \sin(2\theta) \cos \theta$.

(l) $\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta$.

(m) $1 + \tan^2 \theta = \sec^2 \theta$.

(n) $1 + \csc^2 \theta = \cot^2 \theta$.

(o) $2 \cos^2(2x) = 2 \sin^4 \theta + 2 \cos^4 \theta - \sin^2(2\theta)$.

(p) $\frac{1 + \tan(\frac{\theta}{2})}{1 - \tan(\frac{\theta}{2})} = \tan \theta + \sec \theta$.

12. Find all values of x in the interval $[0, 2\pi]$ that satisfy the equation.

(a) $2 \cos x - 1 = 0$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

(b) $\sin(2x) = \cos x$.

$$\frac{9}{10} = x \text{ or } \frac{9}{10} = x$$

(c) $\sqrt{3} \sin x = \sin(2x)$.

$$\frac{9}{10} = x \text{ or } \frac{9}{10} = x$$

(d) $2 \sin^2 x = 1$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

(e) $2 + \cos(2x) = 3 \cos x$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

(f) $2 \cos x + \sin(2x) = 0$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

(g) $2 \cos^2 x - (1 + \sqrt{2}) \cos x + \frac{\sqrt{2}}{2} = 0$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

(h) $|\tan x| = 1$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

(i) $3 \cot^2 x = 1$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

(j) $\sin x = \tan x$.

$$\frac{x}{2} = x \text{ or } \frac{x}{2} = x$$

Solution. 12.g Set $\cos x = u$. Then

$$2 \cos^2 x - (1 + \sqrt{2}) \cos x + \frac{\sqrt{2}}{2} = 0$$

becomes

$$2u^2 - (1 + \sqrt{2})u + \frac{\sqrt{2}}{2} = 0.$$

This is a quadratic equation in u and therefore has solutions

$$\begin{aligned} u_1, u_2 &= \frac{1 + \sqrt{2} \pm \sqrt{(1 + \sqrt{2})^2 - 4\sqrt{2}}}{2} \\ &= \frac{1 + \sqrt{2} \pm \sqrt{1 - 2\sqrt{2} + 2}}{2} \\ &= \frac{1 + \sqrt{2} \pm \sqrt{(1 - \sqrt{2})^2}}{2} \\ &= \frac{1 + \sqrt{2} \pm (1 - \sqrt{2})}{2} = \begin{cases} \frac{1}{2} & \text{or} \\ \frac{\sqrt{2}}{2} \end{cases} \end{aligned}$$

Therefore $u = \cos x = \frac{1}{2}$ or $u = \cos x = \frac{\sqrt{2}}{2}$, and, as x is in the interval $[0, 2\pi]$, we get $x = \frac{\pi}{3}, \frac{5\pi}{3}$ (for $\cos x = \frac{1}{2}$) or $x = \frac{\pi}{4}, \frac{7\pi}{4}$ (for $\cos x = \frac{\sqrt{2}}{2}$).

13. Evaluate the limits. Justify your computations.

$$(a) \lim_{x \rightarrow 2} 2x^2 - 3x - 6.$$

$$(b) \lim_{x \rightarrow -1} \frac{x^4 - x}{x^2 + 2x + 3}.$$

$$(c) \lim_{x \rightarrow -1} \frac{1}{x^2 - 3x + 2}.$$

$$(d) \lim_{x \rightarrow -2} \sqrt{x^4 + 16}.$$

$$(e) \lim_{x \rightarrow 8} (1 + \sqrt[3]{x})(2 - x).$$

14. Evaluate the limit if it exists.

$$(a) \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x - 2}.$$

$$(b) \lim_{x \rightarrow 3} \frac{x^2 - 3x}{x^2 - 2x - 3}.$$

$$(c) \lim_{x \rightarrow -2} \frac{2x^2 + x - 6}{x^2 - 4}$$

$$(d) \lim_{x \rightarrow 2} \frac{x^2 - 5x - 6}{x - 2}.$$

$$(e) \lim_{x \rightarrow -1} \frac{x^2 - 3x}{x^2 - 2x - 3}.$$

$$(f) \lim_{x \rightarrow -2} \frac{x^2 - 4}{2x^2 + 5x + 2}.$$

$$(g) \lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{3x^2 - 2x - 5}.$$

$$(h) \lim_{x \rightarrow -4} \frac{x^2 + 7x + 12}{x^2 + 6x + 8}.$$

$$(i) \lim_{h \rightarrow 0} \frac{(-3 + h)^2 - 9}{h}.$$

$$(j) \lim_{h \rightarrow 0} \frac{(-2 + h)^3 + 8}{h}.$$

$$(k) \lim_{x \rightarrow -3} \frac{x + 3}{x^3 + 27}.$$

$$(l) \lim_{x \rightarrow 1} \frac{x^4 - 1}{x^3 - 1}.$$

$$(m) \lim_{h \rightarrow 0} \frac{\sqrt{4 + h} - 2}{h}.$$

$$(n) \lim_{x \rightarrow 3} \frac{\sqrt{5x + 1} - 4}{x - 3}.$$

$$(o) \lim_{x \rightarrow -3} \frac{\sqrt{x^2 + 16} - 5}{x + 3}.$$

$$(p) \lim_{x \rightarrow -3} \frac{\frac{1}{3} + \frac{1}{x}}{3 + x}.$$

$$(q) \lim_{x \rightarrow -2} \frac{x^2 + 4x + 4}{x^4 - 16}.$$

$$(r) \lim_{x \rightarrow 0} \frac{\sqrt{1 + x} - \sqrt{1 - x}}{x}.$$

$$(s) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x^2 + x} \right).$$

$$(t) \lim_{x \rightarrow 9} \frac{3 - \sqrt{x}}{9x - x^2}.$$

$$(u) \lim_{h \rightarrow 0} \frac{(2 + h)^{-1} - 2^{-1}}{h}.$$

$$(v) \lim_{x \rightarrow 0} \left(\frac{1}{x\sqrt{1 + x}} - \frac{1}{x} \right).$$

$$(w) \lim_{h \rightarrow 0} \frac{(x + h)^3 - x^3}{h}.$$

$$(x) \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h}.$$

$$(y) \lim_{h \rightarrow 0} \frac{\frac{1}{(2+h)^2} - \frac{1}{4}}{h}.$$

$$(z) \lim_{h \rightarrow 0} \frac{\frac{1}{(1+h)^2} - 1}{h}.$$

Solution. 14.a

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 3)(\cancel{x - 2})}{\cancel{x - 2}} \quad \left| \begin{array}{l} \text{factor and cancel} \end{array} \right. \\ &= 2 - 3 = -1 \end{aligned}$$

Solution. 14.c

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{2x^2 + x - 6}{x^2 - 4} &= \lim_{x \rightarrow -2} \frac{(2x - 3)(\cancel{x + 2})}{(x - 2)(\cancel{x + 2})} \quad \left| \begin{array}{l} \text{factor and cancel} \end{array} \right. \\ &= \frac{(2(-2) - 3)}{-2 - 2} \quad \left| \begin{array}{l} \text{substitute} \end{array} \right. \\ &= \frac{7}{4} \end{aligned}$$

Solution. 14.f

$$\begin{aligned}\lim_{x \rightarrow -2} \frac{x^2 - 4}{2x^2 + 5x + 2} &= \lim_{x \rightarrow -2} \frac{(x-2)\cancel{(x+2)}}{(2x+1)\cancel{(x+2)}} \quad \left| \text{factor and cancel} \right. \\ &= \frac{(-2) - 2}{2(-2) + 1} = \frac{4}{-3}.\end{aligned}$$

Solution. 14.g

$$\begin{aligned}\lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{3x^2 - 2x - 5} &= \lim_{x \rightarrow -1} \frac{(2x+1)\cancel{(x+1)}}{(3x-5)\cancel{(x+1)}} \quad \left| \text{factor and cancel} \right. \\ &= \frac{2(-1) + 1}{3(-1) - 5} = \frac{1}{-8}.\end{aligned}$$

Solution. 14.h

$$\begin{aligned}\lim_{x \rightarrow -4} \frac{x^2 + 7x + 12}{x^2 + 6x + 8} &= \lim_{x \rightarrow -4} \frac{(x+3)\cancel{(x+4)}}{(x+2)\cancel{(x+4)}} \quad \left| \text{factor} \right. \\ &= \frac{-4 + 3}{-4 + 2} = -\frac{1}{2}.\end{aligned}$$

Solution. 14.x

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h} &= \lim_{h \rightarrow 0} \frac{x^2 - (x+h)^2}{hx^2(x+h)^2} = \lim_{h \rightarrow 0} \frac{x^2 - (x^2 + 2xh + h^2)}{hx^2(x+h)^2} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(-2x+h)}{\cancel{h}x^2(x+h)^2} = \frac{-2x+0}{x^2(x+0)^2} = -\frac{2}{x^3}.\end{aligned}$$

Solution. 14.y.

Variant I.

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\frac{1}{(2+h)^2} - \frac{1}{4}}{h} &= \lim_{h \rightarrow 0} \frac{\frac{4-(2+h)^2}{4(2+h)^2}}{h} \\ &= \lim_{h \rightarrow 0} \frac{4 - (4 + 4h + h^2)}{4h(2+h)^2} \\ &= \lim_{h \rightarrow 0} \frac{-4h - h^2}{4h(2+h)^2} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(-4-h)}{4\cancel{h}(2+h)^2} \quad \left| \text{substitute } h = 0 \right. \\ &= \frac{-4-0}{4(2+0)^2} \\ &= -\frac{1}{4}\end{aligned}$$

Variant II.

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\frac{1}{(2+h)^2} - \frac{1}{4}}{h} &= \frac{d}{dx} \left(\frac{1}{x^2} \right) \Big|_{x=2} \\ &= \left(\frac{-2}{x^3} \right) \Big|_{x=2} \\ &= -\frac{1}{4}\end{aligned}$$

Solution. 14.z.

Variant I.

$$\begin{aligned}
\lim_{h \rightarrow 0} \frac{\frac{1}{(1+h)^2} - 1}{h} &= \lim_{h \rightarrow 0} \frac{\frac{1-(1+h)^2}{(1+h)^2}}{h} \\
&= \lim_{h \rightarrow 0} \frac{1 - (1 + 2h + h^2)}{h(1+h)^2} \\
&= \lim_{h \rightarrow 0} \frac{-2h - h^2}{h(1+h)^2} \\
&= \lim_{h \rightarrow 0} \frac{\cancel{h}(-2 - h)}{\cancel{h}(1+h)^2} \quad \left| \text{substitute } h = 0 \right. \\
&= \frac{-2 - 0}{(1+0)^2} \\
&= -2.
\end{aligned}$$

Variant II.

$$\begin{aligned}
\lim_{h \rightarrow 0} \frac{\frac{1}{(1+h)^2} - 1}{h} &= \frac{d}{dx} \left(\frac{1}{x^2} \right) \Big|_{x=1} \quad \left| \text{derivative definition} \right. \\
&= \left(\frac{-2}{x^3} \right) \Big|_{x=1} \\
&= -2.
\end{aligned}$$